D.A.K.S.H. SYSTEM PHASING MATRIX

End-to-End Tactical Implementation Guide across 6 Sequential Engineering Phases

PHASE 1: LOCAL CORE & TEMPORAL ENGINE (arch/core-engine)

Primary Objective: Build the deterministic algorithmic core: local persistence, mathematical modulo scheduling for alternate days, and midnight debt rollover.

- **Subphase 1.1 (Encrypted Persistence Schema):** SQLite / Room database modeling tasks, cadence rules, rollover ledgers, and reversible action audit trails.
- **Subphase 1.2 (Alternating Modulo Arithmetic):** Anchor-based modulo parity calculation (DayCount % 2) guaranteeing absolute synchronization for alternate-day habits (shampoo, workout splits).
- **Subphase 1.3 (Midnight Rollover Daemon):** Automated 23:59:59 rollover evaluator carrying uncompleted tasks to tomorrow with #rolled_over tags and debt counters.
- **Subphase 1.4 (Thread-Safe Event Bus):** In-process asynchronous event bus dispatching intent, status, and telemetry signals.

PHASE 2: ANDROID NATIVE SENTINEL BRIDGE (platform/android-sentinel)

Primary Objective: Grant D.A.K.S.H. sovereign device control via native Android APIs and explicit user authorization protocol.

- **Subphase 2.1 (Consent & Protocol Gatekeeper):** Interactive high-privilege permission onboarding (Contacts, NotificationListener, Exact Alarms, Telephony).
- **Subphase 2.2 (Native Contacts Controller):** ContentResolver bridge to query, match, and rename contacts (e.g., 'Rename Priya to Swimi') with reversible state snapshots.
- **Subphase 2.3 (Ambient Notification Interceptor):** NotificationListenerService extracting actionable items (WhatsApp, bank OTPs, delivery alerts) while filtering marketing spam.
- **Subphase 2.4 (Doze & Battery Resilience):** WorkManager and foreground execution pipeline allowing morning briefings to fire through deep battery sleep.

PHASE 3: TACTICAL INTENT ROUTER & MULTI-AGENT BRAIN (agent/cognitive-brain)

Primary Objective: Sub-100ms multi-model intent routing, parameter validation, and tactical persona synthesis.

- **Subphase 3.1 (Sub-100ms Router):** Fast SLM parsing voice inputs into structured JSON tool calls (Device Mutation, Temporal Operation, Query).
- **Subphase 3.2 (Action Dispatcher & Validator):** Sanitizes parameters (phone numbers, contact names, exact times) before native execution.
- **Subphase 3.3 (Tactical Persona Engine):** Paul Bettany / Vision-inspired cadence generator delivering concise, authoritative briefings without AI boilerplate.

PHASE 4: BARITONE VOICE & FULL-DUPLEX AUDIO (voice/neural-acoustic)

Primary Objective: Cinematic acoustic resonance, custom wake-word, and real-time barge-in capability.

- **Subphase 4.1 (Streaming ASR & Wake-Word):** Instant 'Hey Daksh' detection and real-time speech tokenization.
- **Subphase 4.2 (Studio-Grade Baritone TTS):** Low-latency neural streaming voice with studio EQ and acoustic depth.
- **Subphase 4.3 (Instant Barge-In Handling):** Hardware microphone interrupt loop immediately halting audio output when user speaks.

PHASE 5: KITE-TACTICAL UI (ui/kite-tactical)

Primary Objective: Dual-theme financial HUD with default paper-white tactical mode and dynamic 3-column ticket ledger.

- **Subphase 5.1 (Light-First Dual Theme):** Paper-white/ivory canvas (#F8FAFC) with 1px architectural lines (#E2E8F0) and Obsidian Dark mode (#0B0D11).

- **Subphase 5.2 (3-Column Execution Ledger):** Live telemetry HUD: Today's Cadences, Rollover Debt Tickets, and Sentinel Notification Stream.
- **Subphase 5.3 (Harmonic Spectrum Visualizer):** Razor-thin 2px soundwave bar reacting in real-time to voice and reasoning state.

PHASE 6: FIELD HARDENING & SOVEREIGN REPO (main)

Primary Objective: 24h battery profiling, offline failover, and transition to sovereign private repository.

- **Subphase 6.1:** 24-hour battery consumption optimization for continuous background notification listening.
- **Subphase 6.2:** Offline on-device failover when network drops.